Lab 7: Fuzzing for Input Validation Bugs

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Repo	L6_L7_03_coffee_shop_order (Lab 7)
Tools	Python 3.12, Hypothesis 6.150.0, pytest 9.1.1

Step 1: Installed dependencies with `pip install hypothesis pytest`.

Step 2: Running the buggy processor.py

```
C:\Users\Mathin M\Desktop\SE-Lab>copy /Y processor_original.py processor.py
        1 file(s) copied.

C:\Users\Mathin M\Desktop\SE-Lab>python processor.py
==== Running lookup_drink_code ====
Enter drink code prefix (e.g. LAT, CAP, ESP): XYZ
Traceback (most recent call last):
  File "C:\Users\Mathin M\Desktop\SE-Lab\processor.py", line 41, in <module>
    main()
  File "C:\Users\Mathin M\Desktop\SE-Lab\processor.py", line 27, in main
    price = lookup_drink_code(drink_input)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\Mathin M\Desktop\SE-Lab\processor.py", line 10, in lookup_drink_code
    return menu_prices[code_prefix]
           ~~~~~~~~~~~^^^^^^^^^^^^^
KeyError: 'XYZ'
```

Input **XYZ** (an unknown drink code) crashes lookup_drink_code with an unhandled **KeyError: 'XYZ'**.

Step 4: Running test_processor.py on the buggy code

```
test_processor.py:75: AssertionError
=========================================================== short test summary info ============================================================
FAILED test_processor.py::test_lookup_drink_code_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_lookup_drink_code_valid_prefix - AssertionError: lookup_drink_code crashed with: KeyError: 'MOC'
FAILED test_processor.py::test_calculate_average_shot_cost_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_calculate_average_shot_cost_csv - AssertionError: calculate_average_shot_cost crashed with: ValueError: could not convert str
ing to float: ''
FAILED test_processor.py::test_parse_order_modifiers_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_parse_order_modifiers_roundtrip - AssertionError: parse_order_modifiers crashed with: ValueError: not enough values to unpack
 (expected 2, got 1)
============================================================ 6 failed in 19.73s ============================================================

C:\Users\Mathin M\Desktop\SE-Lab>
```

Hypothesis found failures in all three functions: KeyError (unknown/None code, 'MOc' key typo), ValueError on empty CSV / non-numeric prices (divide-by-zero risk), and ValueError on modifier pairs without ':'. **6 failed**.

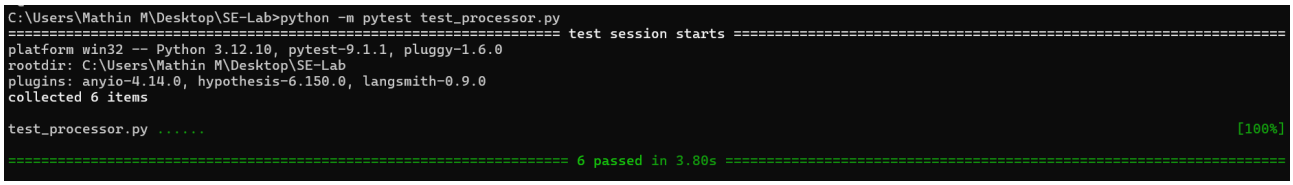
Step 5: Running the fixed processor.py

```
C:\Users\Mathin M\Desktop\SE-Lab>python processor.py
==== Running lookup_drink_code ====
Enter drink code prefix (e.g. LAT, CAP, ESP): XYZ
Invalid or unknown drink code.

==== Running calculate_average_shot_cost ====
Enter CSV of shot prices (e.g. 1.0, 1.5, 2.0):
```

Same input **XYZ** is now handled gracefully: "Invalid or unknown drink code.", and the program continues instead of crashing.

Step 6: Re-running test_processor.py on the fixed code



All fuzz tests pass against the hardened functions: **6 passed**.

Bugs found and fixes applied

Function	Bug found by fuzzing	Fix
lookup_drink_code	None input → AttributeError; unknown code → KeyError; key 'MOc' never matches after .upper()	Type check, strip, dict.get() returns None for unknown; key corrected to 'MOC'
calculate_average_shot_cost	'' / non-numeric → ValueError; empty list → ZeroDivisionError; None, nan, inf, negatives accepted	Type check, skip empty items, reject non-numeric / non-finite / negative values, return None if no prices
parse_order_modifiers	Item without ':' or empty string → ValueError; 'a:b:c' → too many values; None → AttributeError	Type check, skip malformed pairs and empty keys, split(':', 1)